

SSH Brute Force Detection using Splunk & Shuffle SOAR - Troubleshooting Guide

This document contains the common issues faced during the project setup and the solutions used to resolve them. The explanations are written in simple language for documentation, viva, and future reference.

1. Splunk Web UI Not Opening

Check whether Splunk service is running using `/opt/splunk/bin/splunk status`. If stopped, start it using `/opt/splunk/bin/splunk start --run-as-root`. Also verify port 8000 is accessible.

2. Universal Forwarder Not Sending Logs

Verify the forward-server configuration and ensure port 9997 is open on the Splunk server. Check Splunk Forwarder status and confirm `auth.log` is being monitored.

3. Logs Not Visible in Splunk

Verify source path `/var/log/auth.log`, check index configuration, and confirm the forwarder is connected.

4. Hydra Showing Connection Refused

SSH was not listening on the default port 22. Investigation using `'ss -tulnp'` showed SSH was running on port 2232.

5. Splunk Alert Not Triggering

Confirm the SPL query returns results. Verify the alert is configured as Real-Time and Trigger = Per-Result.

6. Shuffle Workflow Not Running

Test the webhook manually using curl. If curl triggers the workflow, the issue is likely on the Splunk alert side.

7. Webhook Counter Increasing but No Action Executed

Verify node connections inside the workflow and ensure actions are properly linked to the webhook trigger.

8. Email Node Stuck or Executing Forever

Review SMTP configuration and network connectivity. Check whether outbound SMTP traffic is allowed.

9. Network Unreachable Error

SMTP traffic was initially blocked. Connectivity was tested using 'nc -vz smtp.gmail.com 587'.

10. SMTP Authentication Required (530 Error)

Gmail rejected authentication. The solution was enabling 2-Factor Authentication and generating a Gmail App Password.

11. Emails Sent to Old Recipients

Workflow changes were not saved. Save the workflow and run a new execution after modifying recipients.

12. Email Not Received

Check Spam/Junk folder, verify recipient address, and confirm Shuffle reports successful email delivery.

13. Firewall Troubleshooting

Verify outbound SMTP port 587 is allowed. Test using nc or telnet to smtp.gmail.com.

14. Webhook Validation

Use curl against the Shuffle webhook URL. Successful execution confirms webhook connectivity.

15. Final Validation Checklist

Generate failed SSH logins, verify logs in Splunk, trigger the alert, confirm Shuffle execution, and verify email delivery.